Final - Probability I (2024-25)
Time: 3 hours.

Attempt all questions. The total marks is 52, and the maximum you can obtain is 50. You may quote any result proved in class without proof.

1. The density function of a random variable X is given by

$$f(x) = \begin{cases} a + bx^2, & 0 \leq x \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

If $E[X] = \frac{3}{5}$, find a and b. **[5 marks]**

2. Consider two special dice, each with *only* 3 faces numbered $\{1, 2, 3\}$. Assume that the three numbers are equally likely to appear when either dice is thrown. Let both the dice be thrown independently, and let X be the score of the first die and Y be the larger of the two scores.

   (a) Write down the joint pmf of X and Y. **[5 marks]**

   (b) Compute $E(X), E(Y), \mathrm{Cov}(X, Y)$. **[5 marks]**

3. Let $X \sim N(5, 9)$ be a Normal random variable with mean 5 and variance 9. Write down $P(4 < X \leq 6)$ in terms of the Standard Normal cumulative distribution function $\Phi(x) = P(Z \leq x)$, where $Z \sim N(0, 1)$. **[5 marks]**

4. Suppose A and B are independently selected random subsets of $\Omega = \{1, 2, \cdots, n\}$ (*including* the empty set and Ω itself as possible subsets). Note that a random subset is obtained by tossing a fair coin independently for each element of $\{1, 2, \cdots, n\}$, and putting the element in the subset if the coin lands heads. Find $P(A \subseteq B)$. **[6 marks]**

5. Prove the following theorem: Let X be a continuous random variable having probability density f_X. Suppose that $g(x)$ is strictly *increasing*, differentiable function of x, with derivative $g'(x)$. Then the random variable defined by $Y = g(X)$ has a probability density given by

$$f_Y(y) = \begin{cases} f_X\left(g^{-1}(y)\right) \cdot \frac{1}{g'(g^{-1}(y))}, & \text{if } y = g(x) \text{ for some } x, \\ 0, & \text{if } y \neq g(x) \text{ for all } x. \end{cases}$$

HINT: Find the CDF first. **[6 marks]**
Use the above theorem to find distribution of $Y = \log X$ where X is an exponential random variable with $\lambda = 1$, that is $f_X(x) = e^{-x}$ for $x > 0$, and $= 0$ otherwise. **[2 marks]**

6. A coin having probability p of coming up heads is continually flipped until *both* heads and tails have appeared.

   (a) Find the expected number of flips. **[6 marks]**

   (b) Find the probability that the last flip lands on heads. **[2 marks]**

7. Let (Ω, P) be a discrete probability space, that is Ω has finite or countably many outcomes.

   (a) Let X and Y be random variables on (Ω, P) such that $X(\omega) \leq Y(\omega)$ for all ω. Show that $E(X) \leq E(Y)$. **[3 marks]**

   (b) Let Z be a *nonnegative* random variable on (Ω, P). Show that for any $t > 0$

$$P(Z \geq t) \leq \frac{E(Z^2)}{t^2}. \quad \textbf{[4 marks]}$$

   HINT: Compare the random variable $t^2 \cdot \mathbf{1}_{\{Z \geq t\}}$ with Z^2. Here

$$\mathbf{1}_{\{Z \geq t\}}(\omega) = \begin{cases} 1, & \text{if } Z(\omega) \geq t, \\ 0, & \text{otherwise.} \end{cases}$$

   (c) For a random variable W on (Ω, P) with $E(W) = \mu$, show that for any $t > 0$

$$P(|W - \mu| \geq t) \leq \frac{\mathrm{Var}(W)}{t^2}. \quad \textbf{[3 marks]}$$